非线性薛定谔方程：

$$i\frac{\partial A}{\partial z} = -\underbrace{\frac{i\alpha}{2}A}_{\text{损耗}} + \underbrace{\frac{\beta_2}{2}\frac{\partial^2 A}{\partial T^2}}_{\text{色散}} - \underbrace{\gamma|A|^2 A}_{\text{非线性}}.$$

色散与非线性何者占主导？

一取决于入射脉宽 T_0，峰值功率 P_0.

引入归一化时间尺度 $\tau = \frac{T}{T_0} = \frac{t-\frac{z}{v_g}}{T_0}$.

归一化振幅 U： $A(z,\tau) = \sqrt{P_0}\exp(-\alpha z/2)U(z,\tau)$. （损耗）

U 满足：（$sgn(\beta_2) = \pm 1$）

$$i\frac{\partial U}{\partial z} = \frac{sgn(\beta_2)}{2L_D}\frac{\partial^2 U}{\partial \tau^2} - \frac{\exp(-\alpha z)}{L_{NL}}|U|^2 U.$$

其中引入了色散长度 L_D & 非线性长度 L_{NL}.

$$L_D = \frac{T_0^2}{|\beta_2|}, \quad L_{NL} = \frac{1}{\gamma P_0}$$

- 二者给出了脉冲演化中，色散与非线性二者谁更重要的长度尺度.

① $L \ll L_D$, $L \ll L_{NL}$. 吸收. 主要作用：传输. 光通信.

② $L \ll L_{NL}$, $L \simeq L_D$. 色散. 条件：

$$L_D / L_{NL} = \frac{\gamma P_0 T_0^2}{|\beta_2|} \ll 1.$$

② $L \ll L_D$, $L \approx L_{NL}$　自相位调制. 频谱变化.

$$L_D / L_{NL} = \frac{\gamma P_0 T_0^2}{|\beta_2|} \gg 1.$$

- 弱群速度色散下，自相位调制 → 脉冲整形. ∴ 若脉冲前、后沿变陡，即使 $L_D / L_{NL} \gg 1$，色散而十分重要.

④ $L \gg L_D$, $L \gg L_{NL}$. 二者共同作用.

- 反常色散 ($\beta_2 < 0$). — Soliton.
- 正常色散 ($\beta_2 > 0$) — SPM & GVD → Compression.

Dispersion only. $\gamma = 0$.

$U(T)$ 满足：

$$i\frac{\partial U}{\partial z} = \frac{\beta_2}{2}\frac{\partial^2 U}{\partial T^2}$$

类似于连续光衍射的傍轴波动方程，∴色散的时间效应与衍射的空间效应类似.

FT 求解：$\tilde{U}(z,\omega) \longleftrightarrow U(z,T)$.

$$U(z,T) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{U}(z,\omega) e^{-i\omega T} d\omega.$$

$\Rightarrow$

$$i\frac{\partial \tilde{U}}{\partial z} = -\frac{1}{2}\beta_2 \omega^2 \tilde{U}$$

解：

$$\tilde{U}(z,\omega) = \tilde{U}(0,\omega)\exp\left(\frac{i}{2}\beta_2\omega^2 z\right).$$

群速度色散改变有个频谱分量的相位.

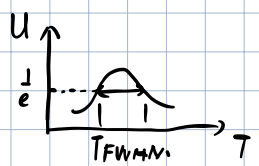
通解：

$$U(z,T) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{U}(0,\omega)\exp\left(\frac{i}{2}\beta_2\omega^2 z - i\omega T\right) d\omega \qquad (3.2.5).$$

Example : Gaussian pulse.

$$U(0,T) = \exp\left(-\frac{T^2}{2T_0^2}\right).$$

T_0 : — $T_{FWHM} = 2\sqrt{\ln 2}\, T_0 \approx 1.665\, T_0$.



$$\tilde{U}(0,\omega) = \int_{-\infty}^{\infty} \exp\left(-\frac{T^2}{2T_0^2}\right)\exp(i\omega T)\,dT = \sqrt{2T_0^2\pi}\,\exp\left(-\frac{\omega^2}{2/T_0^2}\right)$$

$$= \sqrt{2T_0^2\pi}\,\exp\left(-\frac{T_0^2\omega^2}{2}\right)$$

$$U(z,T) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \sqrt{2\pi T_0^2}\,\exp\left(-\frac{T_0^2\omega^2}{2}\right)\exp\left(\frac{i}{2}\beta_2\omega^2 z - i\omega T\right)d\omega.$$

$$= \frac{T_0}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \exp\left[\left(-\frac{T_0^2}{2} + \frac{i}{2}\beta_2 z\right)\omega^2 - i\omega T\right]d\omega.$$

$$= \frac{T_0}{\sqrt{2\pi}}\,\frac{\sqrt{\pi}}{\sqrt{\frac{T_0^2}{2} - \frac{i}{2}\beta_2 z}}\,\exp\left(-\frac{T^2}{2T_0^2 - i2\beta_2 z}\right)$$

$U(0,T)$ $\longrightarrow$ $U(z,T)$

$\exp\left(-\frac{T^2}{2T_0^2}\right)$ $\qquad$ $\frac{T_0}{\sqrt{T_0^2 - i\beta_2 z}}\exp\left(-\frac{T^2}{2T_0^2 - i2\beta_2 z}\right)$

- Gaussian pulse 在传输过程中保持形状不变，但宽度 T_1 ↗.

$$T_1(z) = T_0\left[1 + (z/L_D)^2\right]^{1/2}.$$

色散长度 $L_D = T_0^2/|\beta_2|$.

- 传输后变为啁啾脉冲.

$$U(z,T) = |U(z,T)| \exp[i\phi(z,T)].$$

$$\phi(z,T) = -\frac{sgn(\beta_2)(z/L_D)}{1+(z/L_D)^2}\frac{T^2}{2T_0^2} + \frac{1}{2}\arctan\left[sgn(\beta_2)\frac{z}{L_D}\right]$$

中心频率 ω_0 两侧:

$$\delta\omega(T) = -\frac{\partial\phi}{\partial T} = \frac{sgn(\beta_2)(z/L_D)}{1+(z/L_D)^2}\frac{T}{T_0^2}$$ 频率线性变化.

- 正常色散: $\beta > 0$ $\delta\omega > 0$, $\delta\omega \nearrow$
- 反常色散: $\beta < 0$ $\delta\omega < 0$ $\delta\omega \searrow$

所有频谱分量同相 $\longrightarrow$ 变换极限.

Gaussian pulse 带初始啁啾.

$$U(0,T) = \exp\left[-\frac{(1+iC)}{2}\frac{T^2}{T_0^2}\right].$$

C: 啁啾参量.

· C可由脉冲谱宽估算:

$$\tilde{U}(0,\omega) = \left(\frac{2\pi T_0^2}{1+iC}\right)^{1/2} \exp\left[-\frac{\omega^2 T_0^2}{2(1+iC)}\right]$$

频谱半高宽度 ($\frac{1}{e}$ 处):

$$\Delta\omega = (1+C^2)^{1/2} / T_0.$$

类似前面地, 利用 FT, 有:

$$U(z,T) = \frac{T_0}{[T_0^2 - i\beta_2 z(1+iC)]^{1/2}} \exp\left[-\frac{(1+iC)T^2}{2[T_0^2 - i\beta_2 z(1+iC)]}\right]$$

Chirped Gaussian pulse 传输中保持其形状.

脉宽

$$T_1/T_0 = \left[\left(1+\frac{C\beta_2 z}{T_0^2}\right)^2 + \left(\frac{\beta_2 z}{T_0^2}\right)^2\right]^{1/2}.$$

啁啾参量

$$C_1(z) = C + (1+C^2)(\beta_2 z / T_0^2).$$

Def 归一化传输距离 $\xi = z/L_D$.